

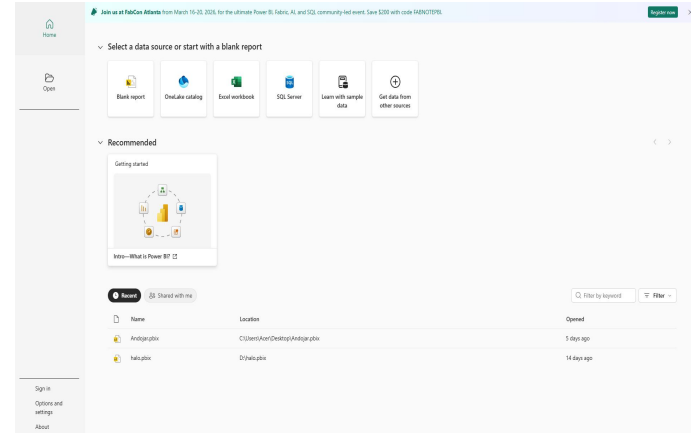
Introduction to Business Intelligence & Power BI

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IS 107 - Business Intelligence

PART 1: Launching Power BI & Loading Data

Step 1: Open Power BI Desktop

- Click Start
- Open Microsoft Power BI Desktop
- Wait for the startup screen



PART 1: Launching Power BI & Loading Data

Step 2: Load the Dataset

1. Click Home Tab

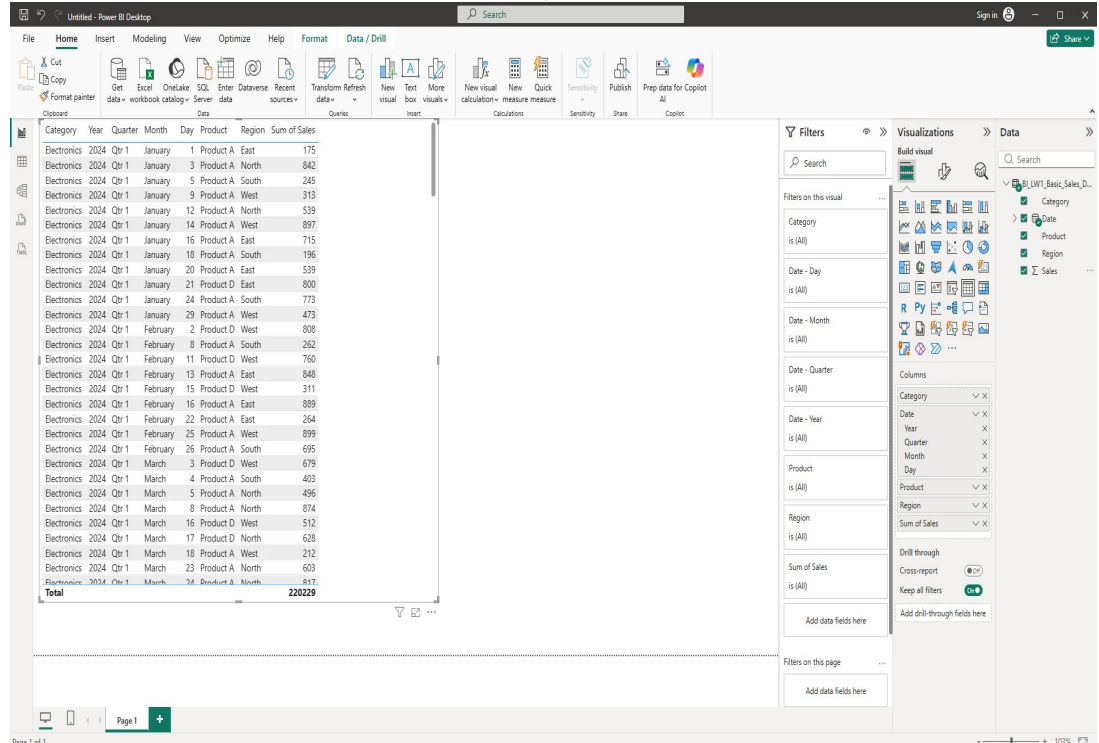
2. Click Get Data

3. Select Text/CSV

4. Browse and select:

Week1_Basic_Sales_Data.csv

5. Click Load



PART 1: Launching Power BI & Loading Data

Step 3: Verify Data in Data View

1. Click the Data View icon (table icon on the left)

2. Check:

- Are all columns visible?
- Is “Date” formatted as Date?
- Is “Sales” formatted as Decimal Number?

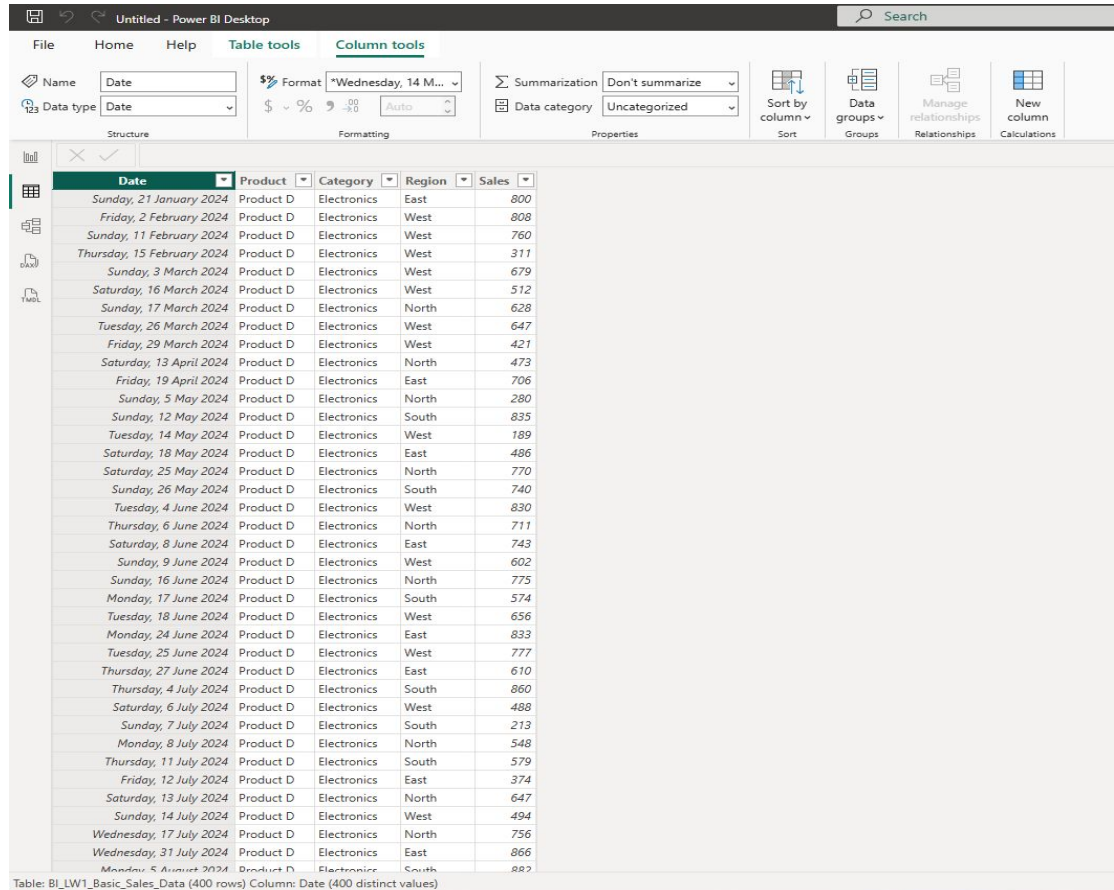
If Data Type is Incorrect:

1. Click the column

2. Go to Column Tools

3. Change Data Type accordingly:

- Date → Date
- Sales → Decimal Number
- Product/Category/Region → Text



Date	Product	Category	Region	Sales
Sunday, 21 January 2024	Product D	Electronics	East	800
Friday, 2 February 2024	Product D	Electronics	West	808
Sunday, 11 February 2024	Product D	Electronics	West	760
Thursday, 15 February 2024	Product D	Electronics	West	311
Sunday, 3 March 2024	Product D	Electronics	West	679
Saturday, 16 March 2024	Product D	Electronics	West	512
Sunday, 17 March 2024	Product D	Electronics	North	628
Tuesday, 26 March 2024	Product D	Electronics	West	647
Friday, 29 March 2024	Product D	Electronics	West	421
Saturday, 13 April 2024	Product D	Electronics	North	473
Friday, 19 April 2024	Product D	Electronics	East	706
Sunday, 5 May 2024	Product D	Electronics	North	280
Sunday, 12 May 2024	Product D	Electronics	South	835
Tuesday, 14 May 2024	Product D	Electronics	West	189
Saturday, 18 May 2024	Product D	Electronics	East	486
Saturday, 25 May 2024	Product D	Electronics	North	770
Sunday, 26 May 2024	Product D	Electronics	South	740
Tuesday, 4 June 2024	Product D	Electronics	West	830
Thursday, 6 June 2024	Product D	Electronics	North	711
Saturday, 8 June 2024	Product D	Electronics	East	743
Sunday, 9 June 2024	Product D	Electronics	West	602
Sunday, 16 June 2024	Product D	Electronics	North	775
Monday, 17 June 2024	Product D	Electronics	South	574
Tuesday, 18 June 2024	Product D	Electronics	West	656
Monday, 24 June 2024	Product D	Electronics	East	833
Tuesday, 25 June 2024	Product D	Electronics	West	777
Thursday, 27 June 2024	Product D	Electronics	East	610
Thursday, 4 July 2024	Product D	Electronics	South	860
Saturday, 6 July 2024	Product D	Electronics	West	488
Sunday, 7 July 2024	Product D	Electronics	South	213
Monday, 8 July 2024	Product D	Electronics	North	548
Thursday, 11 July 2024	Product D	Electronics	South	579
Friday, 12 July 2024	Product D	Electronics	East	374
Saturday, 13 July 2024	Product D	Electronics	North	647
Sunday, 14 July 2024	Product D	Electronics	West	494
Wednesday, 17 July 2024	Product D	Electronics	North	756
Wednesday, 31 July 2024	Product D	Electronics	East	866
Monday, 5 August 2024	Product D	Electronics	South	882

PART 2: Exploring the Interface

View	Purpose
Report View	Create dashboards and visuals
Data View	See raw data
Model View	Manage relationships

Switch back to Report View.

Report view

The screenshot shows the Power BI Report View interface. The main area displays a table of sales data with columns: Region, Year, Quarter, Month, Day, Product, and Sum of Sales. The data is filtered by Region (East) and Year (2024). The table shows sales data for the first quarter of 2024, with a total sum of sales of 228229.

Region	Year	Quarter	Month	Day	Product	Sum of Sales
Electronics	2024	Q1	January	1	Product A	135
Electronics	2024	Q1	January	3	Product A	842
Electronics	2024	Q1	January	5	Product A	245
Electronics	2024	Q1	January	9	Product A	315
Electronics	2024	Q1	January	12	Product A	539
Electronics	2024	Q1	January	14	Product A	2492
Electronics	2024	Q1	January	16	Product A	715
Electronics	2024	Q1	January	18	Product A	788
Electronics	2024	Q1	January	20	Product A	539
Electronics	2024	Q1	January	21	Product D	899
Electronics	2024	Q1	January	24	Product A	771
Electronics	2024	Q1	January	28	Product A	475
Electronics	2024	Q1	February	2	Product D	609
Electronics	2024	Q1	February	8	Product A	262
Electronics	2024	Q1	February	11	Product D	769
Electronics	2024	Q1	February	13	Product A	948
Electronics	2024	Q1	February	15	Product D	3715
Electronics	2024	Q1	February	16	Product A	889
Electronics	2024	Q1	February	22	Product A	264
Electronics	2024	Q1	February	25	Product A	889
Electronics	2024	Q1	February	26	Product A	695
Electronics	2024	Q1	March	3	Product D	679
Electronics	2024	Q1	March	4	Product A	481
Electronics	2024	Q1	March	5	Product D	496
Electronics	2024	Q1	March	8	Product A	654
Electronics	2024	Q1	March	10	Product D	512
Electronics	2024	Q1	March	17	Product D	629
Electronics	2024	Q1	March	18	Product A	2162
Electronics	2024	Q1	March	23	Product A	603
Electronics	2024	Q1	March	30	Product A	603
Total						228229

Data view

The screenshot shows the Power BI Data View interface. The main area displays a table of sales data with columns: Date, Product, Region, and Sales. The data is filtered by Date (2024-01-01 to 2024-03-31) and Region (East). The table shows sales data for the first quarter of 2024, with a total sum of sales of 228229.

Date	Product	Region	Sales
2024-01-01	Product A	East	889
2024-01-03	Product A	East	788
2024-01-05	Product A	East	245
2024-01-09	Product A	East	315
2024-01-12	Product A	East	539
2024-01-14	Product A	East	2492
2024-01-16	Product A	East	715
2024-01-18	Product A	East	788
2024-01-20	Product A	East	539
2024-01-21	Product D	East	899
2024-01-24	Product A	East	771
2024-01-28	Product A	East	475
2024-01-30	Product D	East	609
2024-02-02	Product A	East	262
2024-02-08	Product D	East	769
2024-02-11	Product A	East	948
2024-02-13	Product D	East	3715
2024-02-16	Product A	East	889
2024-02-22	Product A	East	264
2024-02-25	Product A	East	889
2024-02-26	Product A	East	695
2024-02-28	Product D	East	679
2024-03-04	Product A	East	481
2024-03-05	Product D	East	496
2024-03-08	Product A	East	654
2024-03-10	Product D	East	512
2024-03-17	Product D	East	629
2024-03-18	Product A	East	2162
2024-03-23	Product A	East	603
2024-03-30	Product A	East	603

Model view

The screenshot shows the Power BI Model View interface. The main area displays a diagram of data relationships. The diagram shows a table named 'Sales' with columns: Date, Product, Region, and Sales. The table is connected to a table named 'Product' with columns: Product, Region, and Sales. The table is also connected to a table named 'Region' with columns: Region and Sales. The diagram shows the relationships between the tables and the columns.

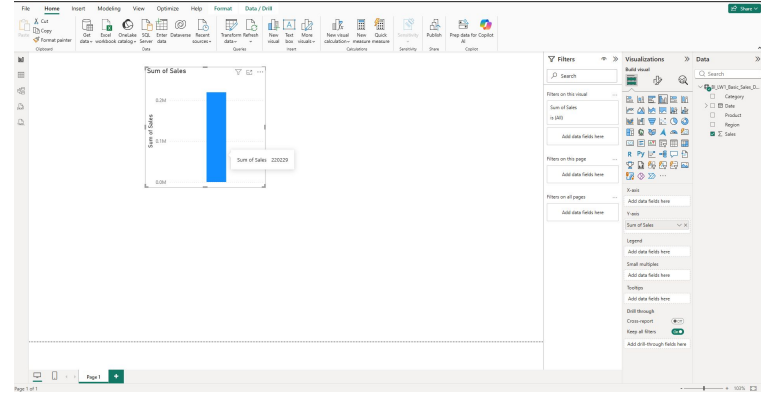
PART 3: Creating Auto-Generated Visuals

Step 1: Quick Visualization

1. Drag Sales into canvas
2. Power BI automatically creates a visual

Question:

- What type of chart was created?
 - The chart shows Sum of Sales $\approx$ 0.2M (200,000+).
- What does it show?
 - Power BI automatically aggregated Sales using SUM, showing the total sales of the dataset.



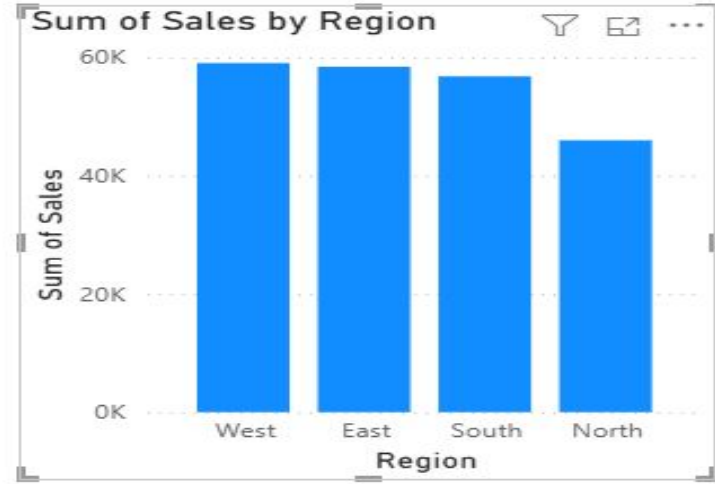
PART 3: Creating Auto-Generated Visuals

Step 2: Create a Sales by Region Chart

1. Click blank canvas
2. Select Clustered Column Chart
3. Drag:
 - Region → X-axis
 - Sales → Values

Question:

- Which region has highest sales?
 - West Region



PART 3: Creating Auto-Generated Visuals

Step 3: Sales by Category

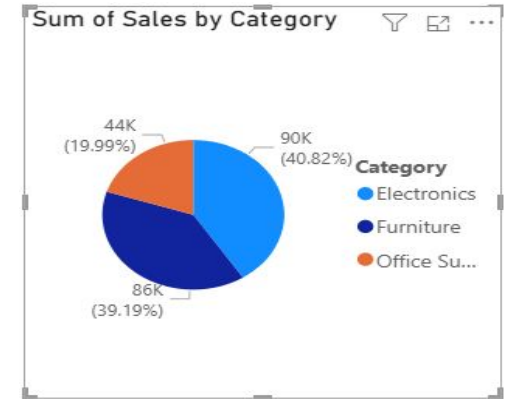
1. Insert a Pie Chart

2. Drag:

- Category → Legend
- Sales → Values

Question:

- Which category dominates?
 - Electronics
- Is the distribution balanced?
 - Mostly yes, but North underperforms compared to others.



PART 3: Creating Auto-Generated Visuals

Step 4: Sales Over Time

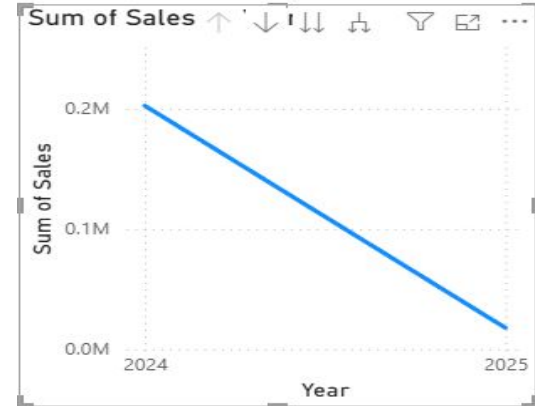
1. Insert Line Chart

2. Drag:

- Date → X-axis
- Sales → Values

Question:

- Is there growth?
 - Nope, the sales go down
- Any noticeable trend?
 - Nope



PART 4: Basic Data Insight Interpretation

Question:

- Which region contributes most revenue?
 - West Region
- Which product category performs best?
 - Electronics
- Are sales consistent across dates?
 - Mostly yes, but North underperforms compared to others.
- What business recommendation can you suggest?
 - Office Supplies

LABORATORY QUESTIONS

Part A – Technical Questions

1. What are the five columns in the dataset?

- Date, Product, Category, Region, Sales

2. What data type is assigned to the “Sales” column?

- Decimal Number. It stores continuous numerical values representing revenue amounts.

3. Which Power BI view allows you to see raw data?

- Data View — the table icon on the left-side navigation bar. It shows every row and column in the loaded dataset.

4. What chart type is best for showing trends over time?

- Line Chart. It maps a continuous axis (Date/Time) on the X-axis, making patterns such as growth or decline immediately visible.

5. What aggregation is automatically applied to Sales?

- SUM (Sum of Sales). Power BI defaults to summing numerical fields when they are dragged onto a visual.

Part B – Analytical Questions

6. Which region has the highest total sales?

- West region. From the sorted bar chart, West has the tallest bar, followed closely by East, then South, with North recording the lowest figures.

7. Which category has the lowest performance?

- Office Supplies. The pie chart shows it accounts for only 19.99% (~44K) of total sales, compared to Electronics (40.82%, ~90K) and Furniture (39.19%, ~86K).

8. Are sales increasing, decreasing, or stable?

- Decreasing over the observed period. The line chart (p. 9) shows a steep downward slope from ~0.2M in 2024 to near zero in 2025, indicating a significant sales drop — likely due to incomplete 2025 data.

9. If you were a manager, which region would you prioritize?

- North region should be prioritized for investigation and improvement, as it is the weakest performer. West and East should be studied as best-practice regions to replicate their success strategies.

10. Provide one actionable recommendation based on the data.

- Reallocate marketing and sales resources from Office Supplies to Electronics and Furniture, which jointly account for ~80% of revenue. In the North region, deploy a targeted sales campaign modeled after the West region's approach.

ENHANCEMENT SECTION

Advanced Exploration

Students must perform the following:

Task 1: Add a Card Visualization

1. Insert Card

2. Drag Sales

3. Format:

- Increase font size
- Change title to “Total Sales”

Question:

- What is the total sales amount?
 - Total Sales amount: 220k



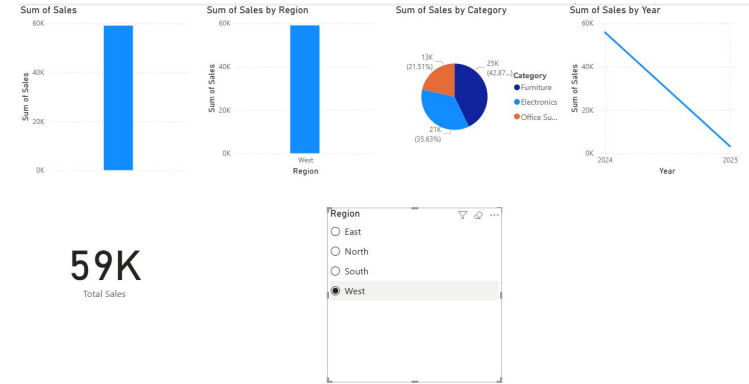
ENHANCEMENT SECTION

Task 2: Add Slicer

1. Insert Slicer
2. Drag Region
3. Test filtering

Question:

- What happens to other visuals when you click a region?
 - All visuals on the page filter to show data ONLY for the selected region.
- Why is filtering important in BI?
 - Filtering is important because it:
 - Allows users to analyze specific segments of data
 - Helps in comparing performance between regions



ENHANCEMENT SECTION

Task 3: Sort Sales

1. Click Region Chart
2. Click three dots (...)
3. Sort by Sales Descending

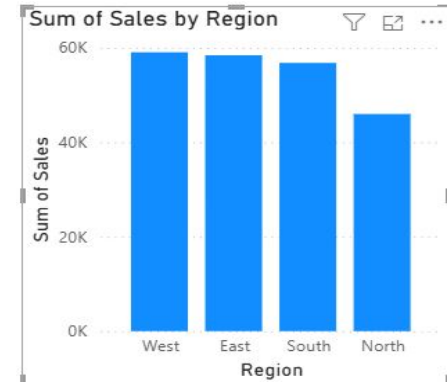
Question:

- Does sorting improve readability?

Yes

- Why?

- sorting dramatically improves readability by pre-attentively ordering information.
- It allows immediate identification of best/worst performers without manual comparison.
- Sorted charts are also easier to annotate (e.g., 'Top performer: West, +28% above average').



ENHANCEMENT SECTION

Task 4: Identify Outliers

- Which region is significantly higher or lower?
- What might explain that difference?
 - North is the clear negative outlier. Based on the sorted bar chart (West → East → South → North), the North column is noticeably shorter than the other three, which are relatively close in height. Possible explanations: fewer sales representatives in the North territory; seasonal demand differences; product-market mismatch where Electronics (the top category) is less in demand in that geography; or simply a smaller target customer population. A manager should drill down into North's product-level data using the Category slicer to isolate whether all categories underperform or just one.